



🎵 Sound Is Vibration 🎵

Physical Science — Sound & Light | Unit 1, Lesson 1 of 3

📖 **Grade:** 1st Grade ⌚ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sound & Light 🖋️ **Unit:** Unit 1, Lesson 1 of 3



Every sound you've ever heard was made by something vibrating — even your own voice!

📋 NGSS Standard:

1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

🎯 Learning Objective

Students will **IDENTIFY** that sound is caused by vibration and **DEMONSTRATE** this by feeling and observing vibrating objects making sound.

🔑 Key Concept

Every sound you've ever heard was made by something **vibrating**.
Vibrations are tiny, rapid back-and-forth movements — too fast to see, but you can feel them!

🔬 The Science

Sound is not magic — it has a physical cause. When an object vibrates (moves rapidly back and forth), it pushes and pulls the air around it. Those pushes and pulls travel through the air as **sound waves** until they reach your ears. No vibration? No sound. It's that simple.

This is why when you strum a guitar string, you can see the blur of the string moving AND hear the note. When you stop the string from moving (by touching it), the vibration stops — and so does the sound, instantly.

Vibration

A rapid back-and-forth movement that creates sound

Pitch

How high or low a sound is (like a mouse squeak vs. a lion roar)



Materials



Household Items to Gather

- Rubber bands — assorted sizes and thicknesses (a bag of mixed sizes works great)
- A shoebox or tissue box (anything with an open top)
- A ruler
- A metal spoon + a pot or metal bowl
- Paper and crayons



No need to buy anything special — grab these from around the house!



Activities



Activity 1 — Feel the Vibration (10 min)

Start with your throat! The throat-humming activity is the most powerful hook — do this one first.

- **Throat hum:** Hum or sing a note. Place two fingers gently on your throat. Feel the buzzing? That's your vocal cords vibrating! Stop humming — does it stop? Yes!
- **Ruler on a table:** Hold a ruler flat on the edge of a table so about half of it sticks out. Hold the other end down firmly. Flick the free end — hear the sound AND watch the blur! That blur IS the vibration.
- **Rubber band twang:** Stretch a rubber band between two fingers. Pluck it with your other hand. See the blur AND hear the twang. Try plucking harder vs. softer. Does the sound change?
- **Bowl buzz:** Hit a metal spoon firmly against the side of a pot or metal bowl. Now IMMEDIATELY touch the bowl gently with your fingertips. Feel the buzz vibrating through the metal? Press harder to stop it — the sound stops too!



After each one, ask: "Where is the vibration coming from? What happens when the vibrating stops?"



Activity 2 — Rubber Band Guitar (15 min)

Build a homemade guitar! Stretch 3–4 rubber bands of different thicknesses around an open shoebox or tissue box.

- **Pluck each one:** Do thick bands sound different from thin ones? Yes! Thick rubber bands vibrate more slowly → lower pitch. Thin ones vibrate faster → higher pitch.
- **Change the length:** Press your finger down on the rubber band between plucks. Does changing the length change the sound? Yes — shorter length = faster vibration = higher pitch. This is exactly how guitar and violin players make different notes!
- **Pluck then grab:** Pluck a rubber band, then immediately press down on it to stop it. What happens to the sound? It stops instantly — because you stopped the vibration!
- **Observe:** Look at the rubber band right after plucking. You can see it blurring back and forth — that is the vibration you are hearing.

 **Connection to real instruments:** Guitars, violins, harps, and pianos all make sound this exact way — vibrating strings! You just built the world's simplest stringed instrument.

Discussion Questions

Question 1

"If vibrations make sound, what do you think happens to the vibrations when you stop a sound — like putting your hand on a ringing bell?"

Suggested answer: The vibrations stop! When you touch the bell, you're stopping it from vibrating, so the sound stops too. Try it right now with your rubber band guitar — pluck it, then grab it. Instant silence!

Question 2

"Can you think of a sound that doesn't come from something vibrating?"

Suggested answer: There isn't one! Every single sound — thunder, music, voices, snapping fingers, a dog barking — is made by vibrations. Challenge them to pick their absolute favorite sound and figure out what is vibrating to make it.

Extensions

-  **Water Ripples:** Fill a bowl with water. Tap the rim gently — watch the ripples! Those ripples are vibrations traveling through water, just like sound travels through air.
-  **Speaker Test:** Put your hand lightly on a speaker or on a phone playing music. Feel it vibrating? That's how speakers make sound — a vibrating cone!
-  **Instrument Hunt:** Look up pictures of a guitar, violin, harp, and piano. What is vibrating in each one? (Strings for all four!)
-  **Read Aloud:** "Sound: Loud, Soft, High, and Low" by Natalie M. Rosinsky, or "What Is Sound?" by Charlotte Guillain
-  **Sing & Feel:** Put your hand on a drum, table, or wooden floor while someone plays music. Can you feel the vibrations in the surface?



Parent/Teacher Notes

- **Start with the throat hum** — it's the most visceral, personal way to feel vibration. Kids are amazed that their own voice is just vibrating muscles in their throat. This hook gets them immediately invested.
- **No materials to buy:** Rubber bands and a shoebox are all you need. Any box with an opening works. Don't have a metal bowl? Any resonant container will do — the vibration is easier to feel in metal, but wood works too.
- **Vocabulary notes:**
 - *Vibration* — the core concept, use it constantly. Every time you hear a sound, ask "what's vibrating?"
 - *Pitch* — high vs. low sound. The rubber band guitar makes this concrete and physical.
 - *Frequency* — just introduce as "how fast something vibrates." No numbers needed at this grade. For curious students, you can mention frequency as a fun fact, but it is not assessed at Grade 1.
- **Why does a thicker rubber band sound lower?** Thicker (or longer) strings vibrate more slowly — that slower speed is a lower frequency, which our ears hear as a lower pitch. Thinner or shorter strings vibrate faster = higher pitch. This is the same physics that determines why a bass guitar sounds deeper than a violin.
- **Connection to music:** Guitars, violins, pianos, and harps all make sound via vibrating strings. Wind instruments use vibrating air columns. Drums use vibrating membrane surfaces. All sound = vibration!
- **Keep it kinesthetic:** First graders learn best by doing and feeling. The activities where they touch the vibrating object (throat, bowl, rubber band) are more valuable than any explanation. Let them explore before you explain.

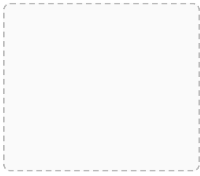
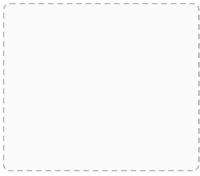
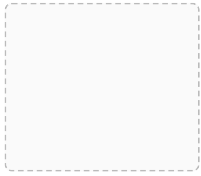
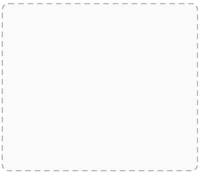
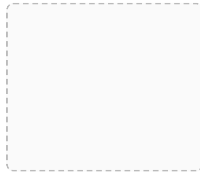
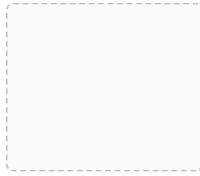
Sound Is Vibration — Worksheet

Little Lab Coats | Grade 1 Physical Science | Unit 1, Lesson 1

Name: _____ Date: _____

Part 1 — Before & After

In each box, draw the object **before** you pluck/hit it on the left. On the right, draw wavy lines to show it **vibrating!**

 Rubber Band		 Ruler		 Metal Bowl	
Before	Vibrating!	Before	Vibrating!	Before	Vibrating!
					

Part 2 — Fill In the Blank

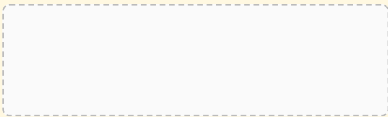
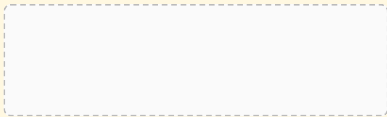
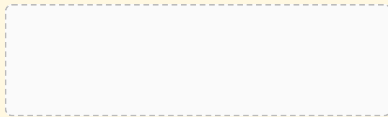
Word Bank: **vibrates** **bounces** **shines**

Sound is made when something _____.

Part 3 — Rubber Band Guitar Record

My Guitar Observations

Draw your three rubber bands below (thin, medium, thick). Then predict which will sound the **lowest** — circle your prediction before you pluck. After testing, circle the one that was actually lowest!

		
Band 1	Band 2	Band 3
Thin / Medium / Thick (circle one)	Thin / Medium / Thick (circle one)	Thin / Medium / Thick (circle one)

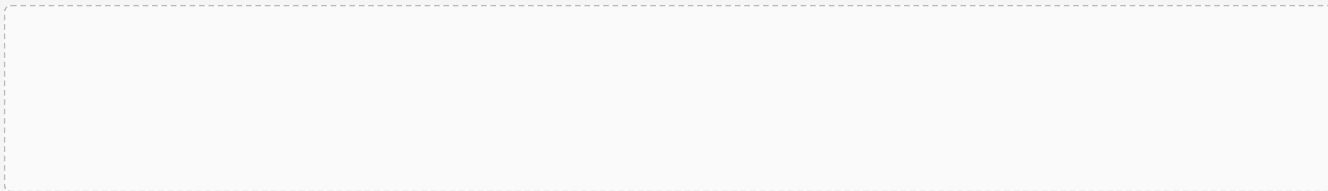
☐ Circle the band that sounded the **lowest** after testing: Band 1 Band 2 Band 3

Part 4 — My Favorite Sound

Write or dictate one sentence:

My favorite sound is _____ and it's made by _____.

 Bonus: Draw your favorite sound in the box below!





PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students



Part 1 — Before & After (Drawing)

- **Before:** The object drawn at rest — a straight rubber band, a flat ruler, a bowl sitting still.
- **Vibrating:** Wavy or zigzag lines coming off the object, or the object shown blurry/jagged to indicate rapid movement. Any representation of back-and-forth motion is correct! There's no single right answer — look for intentional wavy/motion lines.



Part 2 — Fill In the Blank

Sound is made when something **vibrates**. ✓

(Accept: "moves," "moves back and forth," "shakes" — but guide them toward the word *vibrates* as the precise scientific term.)



Part 3 — Rubber Band Guitar Record

- **Which sounds lowest?** The *thickest* rubber band. A thicker (or longer) rubber band vibrates more slowly → lower frequency → lower pitch sound.
- **Which sounds highest?** The *thinnest* rubber band. Thinner strings vibrate faster → higher pitch.
- **What to look for in drawings:** Any representation of three bands of different widths. The key observation is that thickness affects the sound.
- **If kids predict incorrectly:** That's perfectly fine and great for science! Use it to discuss: "What did we predict? What actually happened? Why might thicker be lower?"



Why thicker = lower: A thicker rubber band has more mass and more tension resistance. It takes more energy to move it quickly, so it vibrates at a slower rate (lower frequency). Our ears perceive slower vibrations as lower-pitched sounds. This is the same reason a bass guitar string (thick) sounds much lower than a high E string (thin).



Part 4 — My Favorite Sound

Any answer is correct as long as the student can identify *what* is vibrating to make that sound. Use the table below as a reference:

Sound	What's vibrating?
Thunder	Rapidly expanding air around a lightning bolt
My voice / singing	Vocal cords in your throat
Guitar / violin / piano	Strings
Drums	The drum skin (membrane)
Clapping	Your hands compressing air together rapidly
Birds chirping	Their syrinx (bird vocal organ)
Snapping fingers	Finger hitting palm, compressing air

Key takeaway: there is NO sound that isn't caused by vibration. If a child challenges this, they've just discovered a great science investigation!

✓ Success Criteria for Lesson 1

- Student can state that sound is caused by vibration (in their own words).
- Student can feel or demonstrate vibration in at least two objects.
- Student can predict (even if incorrectly) that thicker rubber bands sound lower — and explain why after testing.
- Student can name a favorite sound and identify what is vibrating to make it.



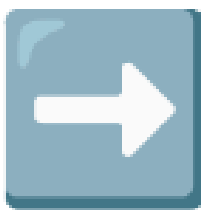
Light Travels in Straight Lines

Physical Science — Sound & Light | Unit 1, Lesson 2 of 3

 **Grade:** 1st Grade

 **Duration:** 30–45 minutes

 **Subject:** Physical Science — Sound & Light



Light always travels in a straight line — it can't bend around corners!

NGSS Standards:

1-PS4-2 — Make observations to construct an evidence-based account that objects in darkness can be seen only when illuminated.

1-PS4-3 — Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light (transparent, translucent, opaque).

Learning Objective

Students will **DEMONSTRATE** that light travels in straight lines and **SORT** materials into *transparent*, *translucent*, and *opaque* categories by testing them with a flashlight.

The Big Idea

Light travels in straight lines — always! It cannot bend around corners or curve through tubes. When something gets in the way of light, a shadow forms in the exact shape of the object. And not all materials treat light the same way: some let all the light through, some let a little through, and some block it completely.



Transparent

Lets ALL light through — you can see clearly. *Example: clear glass, plastic wrap*



Translucent

Lets SOME light through — blurry, not clear. *Example: tissue paper, wax paper*



Opaque

Blocks ALL light — makes a dark shadow. *Example: cardboard, foil, your hand*



Key vocabulary: transparent, translucent, opaque, shadow, beam, reflect, straight



Materials



Gather Before You Start

-  Flashlight
-  3 toilet paper rolls + tape (save these ahead of time!)
- Clear plastic wrap or a clear plastic bag
- Wax paper
- Tissue paper
- Aluminum foil
- A piece of cardboard or a book
- A scrap of thin fabric
- White plastic bag (if available)
- Clear glass or plastic cup



For the Worksheet

- Printed Student Worksheet (last pages of this lesson)
- Crayons or pencil



Let's Get Started

Activity 1 — Does Light Travel Straight?

Step 1 — Flashlight Beam (3 min)

Dim or darken the room. Shine the flashlight across the room toward a wall.

Ask: *"Do you see the beam of light bending or curving anywhere?"*

- No! It goes straight, all the way to the wall — it never curves.

Step 2 — The Toilet Paper Roll Tunnel (10 min)

Line up 3 toilet paper rolls end to end. Use tape to hold them in a straight line.

1. Shine the flashlight in one end — watch light come out the other side! 
2. Now gently **bend the line** slightly (so it's no longer straight). Shine the flashlight again.
Ask: *"Does the light still come out the other end?"*

No! Light can't bend around corners, so it gets blocked inside the bent tube.

This is proof that light travels in **straight lines only**!

Step 3 — Shadow Puppets! (5 min)

Hold your hand between the flashlight and a wall in a dark room. Make a hand shadow!

Ask: "Why does the shadow look exactly like your hand?"

- Because light travels straight — your hand blocks the straight-moving light, so a dark patch the exact shape of your hand appears on the wall.
- If light could bend around corners, there'd be no shadows at all!

Activity 2 — Light Sorting Station

Step 4 — Test Each Material (10–15 min)

In a dim room, hold each material up in front of the flashlight beam. Test them one at a time.

- **Can you see clearly through it?** → **Transparent**
- **Light comes through but it's blurry?** → **Translucent**
- **Blocks the light completely?** → **Opaque**

Sort the materials into three groups as you test them. Fill in the sorting table on the worksheet!

 **Pro tip:** Closing the blinds during the daytime works great — you don't need a fully dark room, just dim enough that you can clearly see the difference between materials.

Discussion Questions

 **Question 1:** "Why do windows need to be transparent rather than opaque? What would happen if all windows were made of cardboard?"

 *We need transparent windows so light can come in — without light during the day, rooms would be completely dark! Cardboard would block all light. We could use opaque walls, but we want windows to let daylight in while keeping weather out.*

 **Question 2:** "When you make a shadow puppet, why does the shadow have the same shape as your hand?"

 *Because light travels in straight lines! Your hand blocks the light, and since light can't bend around it, a dark patch the exact shape of your hand appears on the wall. If light could bend around corners, there'd be no shadows.*

Extensions

-  **Mirror Redirect:** Hold a small mirror in the flashlight beam — can you redirect the light spot to a different spot on the wall? Mirrors *reflect* light, changing its direction (while still traveling straight after the bounce)!
-  **Real-World Connections:** Sunglasses are tinted but still *transparent* — you can see clearly through them, though the tint reduces brightness. Sunblock reflects and absorbs UV light so it doesn't reach your skin.
-  **Read Aloud:** "Light: Shadows, Mirrors, and Rainbows" by Natalie Rosinsky or "What Is Light?" by Ann Coan



Parent/Teacher Notes

- **All household materials:** No purchases needed! Toilet paper rolls are the only "prep" — just save 3 of them ahead of time.
- **Dim the room:** Darkening helps a lot for the sorting activity — closing blinds during the day is enough. A darker room makes the difference between transparent, translucent, and opaque much more obvious.
- **Why can't light bend around corners?** Light is a wave of energy that travels in straight lines — obstacles that block it create shadows. Lenses and mirrors can redirect light, but that's a different mechanism (refraction and reflection). For 1st grade, just emphasize: light goes straight, can't bend around things.
- **Vocabulary to reinforce:** transparent, translucent, opaque, shadow, beam, reflect
- **Tricky case — wax paper:** Wax paper lets a lot of light through but diffuses it (blurry). It's a great example of translucent. Kids may debate transparent vs. translucent — that's great science talk!
- **Assessment:** Can each child correctly sort at least 5–6 materials? Can they say in their own words why the bent tube blocks light? That's success for 1st grade.



Student Worksheet — Light Travels in Straight Lines

Name: _____ Date: _____

Part 1 — Draw the Light!

Draw arrows to show where the light goes in each tube. Does it come out the other end?

Straight Tube



Three rolls in a straight line

Draw arrows showing where the light goes →

Does light come out the end? **Yes / No**

Bent Tube



Three rolls with a bend in the middle

Draw arrows showing where the light goes →

Does light come out the end? **Yes / No**

Part 2 — Fill In the Blank

Light travels in _____ lines.

Part 3 — Light Sorting Table

Test materials with your flashlight. Write or draw 2 items in each column!

 TRANSPARENT Light passes through clearly	 TRANSLUCENT Light comes through — but blurry	 OPAQUE Blocks all light
Write or draw 2 items: 	Write or draw 2 items: 	Write or draw 2 items:

Part 4 — Shadow Time! 🖐️

Trace your hand in the box below. Then draw where the shadow would appear on the wall to the right!

Your hand (trace it here!):

Trace your hand here 🖐️



The wall — draw the shadow:

Draw the shadow here 🖐️

Why does the shadow look like your hand? Because light travels in _____ lines!



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1 — Tube Diagram Answers

- **Straight tube:** Arrows go straight through all three rolls → light comes out the other end. Answer: **Yes**
- **Bent tube:** Arrow enters the first roll, hits the side of the bent tube and stops (blocked). Answer: **No** — light can't bend around corners!

Part 2 — Fill in the Blank

Light travels in *straight* lines.

Part 3 — Light Sorting Table



TRANSPARENT

- ✓ Clear plastic wrap
- ✓ Clear glass/plastic cup
- ✓ Clear plastic bag
- ✓ Clean window glass



TRANSLUCENT

- ✓ Tissue paper
- ✓ Wax paper
- ✓ White/frosted plastic bag
- ✓ Thin fabric
- ✓ Frosted tape



OPAQUE

- ✓ Cardboard
- ✓ Aluminum foil
- ✓ A book
- ✓ Your hand
- ✓ Construction paper

Part 4 — Shadow Drawing

- The shadow on the wall should be the **same shape** as the traced hand — possibly slightly larger (shadows get bigger as the object moves closer to the light source).
- Fill-in answer: "**straight**"



Discussion Notes

- **Question 1 (windows):** Transparent windows let daylight in. Opaque cardboard would block all light — rooms would be completely dark during the day. Accept any answer that shows understanding that transparent materials allow light to pass through.
- **Question 2 (shadow puppets):** Light travels in straight lines, so the hand blocks the light and casts a shadow that matches the hand's exact shape. If light bent around objects, there would be no shadows. Accept any reasoning that connects straight-line travel to shadow shape.
- **Tricky case — wax paper:** Some kids may call it transparent because a lot of light comes through. The key distinction: can you see a clear image through it? No — it's blurry. That's translucent. Celebrate the discussion!



What Success Looks Like

- Child can explain that light travels in a straight line and cannot bend around corners.
- Child can correctly sort at least 5 out of 6+ tested materials into transparent / translucent / opaque.
- Child can connect shadow shape to straight-line light travel.



Mirrors, Signals & Sending Messages



Physical Science — Sound & Light | Unit 1, Lesson 3 of 3



Grade: 1st Grade



Duration: 30–45 minutes



Subject: Physical Science — Sound & Light



Unit 1, Lesson 3 of 3



Mirrors flashing in the sun, drums echoing through the forest — humans have always found ways to talk across distances. Today, you design your own!



NGSS Standard: 1-PS4-4 — Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance.



Learning Objective

Students will **DESIGN** and **TEST** a simple device that uses light or sound to communicate a message over a distance.



The Big Idea

Humans have **always** used light and sound to communicate — from drum messages beating across forests to mirrors flashing signals across mountains. Long before phones or radios existed, people were clever engineers, using the tools they had to send messages to faraway places.

Today, your little scientist gets to be that engineer. They'll choose a signaling device, design a simple code, and test whether their message actually makes it across the room!



Key Vocabulary: communicate, signal, vibration, reflect, code, design



The Science

Sound travels through vibrations. When you speak into a cup connected by a tight string, your voice makes the cup vibrate — and those vibrations travel down the string to the other cup. The tighter the string, the better the vibrations travel!

Light travels in straight lines. When light hits a shiny surface like a mirror or smooth foil, it bounces (reflects) in a new direction. We can aim that reflected light like a pointer to send a signal.

Engineers use these properties of sound and light all the time — from radio towers to fiber-optic cables to satellite dishes. But the same physics that powers the International Space Station's communication system also powers a pair of paper cups and a string!

 **Real-world connections:** Morse code (sound), semaphore flags (light/position), Native American smoke signals, lighthouse beams, ship signal flags, radio waves (invisible light waves that carry our phone calls and Wi-Fi)

Materials

Option A: Mirror Signal

- Small mirror *or* piece of aluminum foil (smoothed flat over cardboard for stiffness)
- Flashlight (optional — sunlight near a window also works great!)

Option B: String Telephone

- 2 paper or plastic cups
- 10–15 feet of string (kite string, yarn, or twine — the thinner the better)
- Sharp pencil or toothpick to poke a small hole in each cup bottom

Option C: Drum Code

- A pot or pan (metal makes the best sound!)
- Wooden spoon or spatula for drumming

Let's Get Started

Step 1 — Hook: What If You Had No Phone? (3–5 min)

Ask: *"Imagine you're camping deep in the woods and you need to tell your friend — who's on the other side of the campsite — that dinner is ready. You can't see them. You have no phone. How would you do it?"*

Let them brainstorm freely. Accept all ideas! Then say: *"Humans have been solving this problem for thousands of years. Today we're going to be the inventors."*

Step 2 — Choose Your Device (2 min)

Present the three options. Let your child (or children) pick one — or try all three if time allows! Each option uses a different scientific principle.

Step 3 — Build It! (5–10 min)

Follow the instructions for the chosen option below. Emphasize: this is **engineering design** — it's okay if it doesn't work perfectly on the first try. That's what iteration is for!

Step 4 — Design Your Code (3 min)

Before testing, agree on a simple code. Examples:

- 1 signal = "Yes" or "Come here"
- 2 signals = "No" or "Wait"
- 3 signals = "I need help"

Write it down on the worksheet so both partners remember it!

Step 5 — Test It! (5–10 min)

Move apart (different sides of the room, or even different rooms for the drum). One person sends a message using only the device. The other tries to receive and understand it. Then switch! Record results on the worksheet.

Step 6 — Improve & Reflect (5 min)

Did it work? What made it harder? What would you change? Complete the worksheet improvement section.



The Three Options



Option A: Mirror Signal

1. Smooth your aluminum foil flat over a piece of cardboard (or use a small mirror). The flatter the surface, the cleaner the reflection!
2. Near a window or with a flashlight, practice bouncing a beam of light onto a wall target. Tilt the mirror until you can see the bright spot appear on the wall.
3. Practice your code: **1 flash = yes, 2 flashes = no, 3 flashes = come here**
4. Have one person turn away (or stand across the room facing away). The "sender" uses flashes to send a message. The "receiver" watches the wall and calls out what they think the message was.
5. Did it work? Try moving farther apart!



Tip: The foil mirror works best when smoothed over a stiff book or piece of cardboard — wrinkles scatter the light beam.

Option B: String Telephone

1. **▲ Adult help needed:** Use a sharp pencil to carefully poke a small hole in the bottom of each cup.
2. Thread one end of the string through the hole in one cup and tie a big knot inside the cup so it can't pull back through. Repeat for the second cup.
3. Both people hold a cup each and walk apart until the string is **very taut** (tight like a guitar string — this is the most important step!).
4. One person speaks softly into their cup. The other holds their cup to their ear. Can they hear?
5. **Experiment:** Now let the string go slack (floppy). Does it still work? Why not?

 **Why it works:** Your voice makes the cup vibrate. Those vibrations travel along the tight string like waves. When the string is slack, the vibrations fizzle out instead of traveling — they need something taut to move through.

Option C: Drum Code

1. Choose a simple code: **1 knock = yes, 2 knocks = no, knock-knock-pause-knock = come here**
2. One person goes to another room (or around the corner). Cover the doorway with a blanket so they absolutely can't see each other — only sound!
3. The "sender" drums their message on the pot. The "receiver" tries to decode it.
4. Switch roles. Can you send a longer message using only your code?
5. **Bonus:** Can you figure out a way to send more complex messages — like "yes-yes-no" or "come here NOW"?

 **Real world:** West African djembe drummers could send messages up to 20 miles away using drum codes that mimicked the tonal patterns of their spoken language!

Discussion Questions

 **Question 1:** "If you were lost in the woods without a phone, what sound or light signal could you use to let rescuers know where you are?"

*Good answers include: yelling, banging rocks together, using a mirror or shiny surface to reflect sunlight, making a fire (smoke by day, fire by night), or blowing a whistle. The international distress signal is **3 of anything** — 3 blasts, 3 flashes, 3 knocks. If they're interested, introduce **SOS** (Save Our Souls) — the international emergency signal that means "I need help!"*

 **Question 2:** "The string telephone only works when the string is tight. What does that tell you about how sound travels?"

Sound needs something to travel through. In a string telephone, the vibrations from your voice travel through the tight string to the other cup. When the string is slack, the vibrations can't travel efficiently — they have nothing to carry them. Sound travels through solids (like string), liquids (like water), and gases (like air) — it needs particles close together to bump into each other and pass the vibration along.



Parent/Teacher Notes

- **This is an engineering design lesson** — there is no "wrong" device. The goal is trying, testing, observing, and improving. Celebrate attempts and iteration. A device that doesn't work the first time is the most educational outcome!
- **String telephone tip:** The string telephone is the most satisfying — kids are always amazed it works. The key is making the string *extremely* taut. A slack string kills the effect every time. Also, speak softly — if you yell, they can just hear you through the air anyway and it defeats the lesson.
- **Foil mirror tip:** Smooth the aluminum foil over a book or piece of stiff cardboard before starting. Wrinkled foil scatters the light beam in too many directions. A smooth surface gives a tight, aimed reflection.
- **Vocabulary to reinforce:** communicate, signal, vibration, reflect, code, design
- **Real-world connections to weave in:** Morse code (dots and dashes = sound), semaphore (light and flag positions), Native American smoke signals, ship signal flags, lighthouse beams. The International Space Station communicates with Earth using *radio waves* — a form of light wave we can't see with our eyes.



Student Worksheet — Lesson 3

Mirrors, Signals & Sending Messages

Name: _____ Date: _____



Design Your Signaling Device

Which device did you choose? Circle one:



Mirror Signal



String Telephone



Drum Code



Draw Your Device & Label Its Parts

Draw your device here. Try to label each part! (e.g., "string," "cup," "mirror," "string knot")



My Signal Code

What code did you and your partner use?

Signal	Meaning
1 signal means:	
2 signals means:	
3 signals means:	



Test Results

I tried to send this message:	My partner received:	Did it work?
		Yes / No

		Yes / No
		Yes / No

 **If I Tried Again, I Would Change...**

 Fill In the Blank

Sound and light can both be used to _____

 **Bonus: Name one way people sent signals before phones were invented!**



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Fill in the blank: "Sound and light can both be used to ____."



communicate (also accept: send messages, talk to people far away, signal)

Discussion Q1: Lost in the woods — what signal could you use?



Good answers: yelling, banging rocks together, using a shiny surface to reflect sunlight toward rescuers, fire/smoke signals, blowing a whistle. **Key concept:** The international distress signal is 3 of anything — 3 blasts, 3 flashes, 3 knocks. SOS = "Save Our Souls" — the universal emergency signal.

Discussion Q2: The string telephone only works when the string is tight. What does that tell us?



Sound needs something to travel through. The vibrations from your voice travel along the tight string to the other cup. When the string is slack, vibrations can't pass through efficiently — they need particles packed together to bump into each other. Sound travels through solids (like string), liquids (like water), and gases (like air).

Bonus: Name one way people sent signals before phones were invented.



Any historically accurate answer: Morse code (telegraph), semaphore flags, smoke signals, drum codes, mirror flashes, lighthouse beams, ship signal flags, town crier bells, carrier pigeons (light signal from a bird's arrival), horn/bugle calls. Praise creative and specific answers!



What Success Looks Like

- Child can name at least one way to use **light** to communicate and one way to use **sound**.
- Child can explain, in their own words, why the string telephone needs a taut string.
- Child completes the engineering design cycle: build → test → observe → improve.
- Child can fill in: "Sound and light can both be used to **communicate**."
- *Bonus:* Child connects today's lesson to at least one real-world historical signaling method.



Vocabulary Check

- **Communicate:** Share information with someone else
- **Signal:** A sound, light, or movement used to send a message
- **Vibration:** A fast back-and-forth movement that creates sound
- **Reflect:** When light bounces off a surface in a new direction
- **Code:** A system where signals stand for words or messages
- **Design:** To plan and create something to solve a problem



🎵 High Sounds, Low Sounds 🎵

Physical Science — Sound & Light | Unit 1, Lesson 4 of 6



Grade: 1st Grade



Duration: 30–45 minutes



Subject: Physical Science — Sound & Light



Different vibrations can make different kinds of sounds — some higher and some lower.



NGSS Standards:

1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.



Learning Objective

Students will **COMPARE** sounds made by different vibrating objects and **DESCRIBE** some sounds as **higher** or **lower** based on how the object is vibrating.



The Big Idea

Lesson 1 showed that sound is made by vibration. In this lesson, students go one step farther: not all vibrations sound the same. When an object vibrates faster, it often makes a higher sound. When it vibrates more slowly, it often makes a lower sound. Children can notice this with rubber bands, rulers, and their own voices.



Key vocabulary: vibrate, pitch, high sound, low sound, tight, loose, fast, slow



Materials



Gather Before You Start

- Empty tissue box or small box with an opening
- 4–6 rubber bands of different sizes or thicknesses
- Pencil or craft stick
- Optional: ruler for a table-edge vibration test



For the Worksheet

- Printed Student Worksheet (last pages of this lesson)
- Pencil or crayons



Let's Get Started

Step 1 — Build a Simple Sound Maker (5 min)

Stretch rubber bands across the opening of a tissue box. Pluck one rubber band and watch it move. Ask: *"What do you see the band doing?"*

Remind students that the band is vibrating, and that vibration makes the sound.

Step 2 — Listen for High and Low (6 min)

Pluck one thin band and then one thicker band. Ask your child to decide which sound seems higher and which seems lower.

Encourage simple language like: *"This one sounds squeaky and high,"* or *"This one sounds deeper and low."*



Helpful setup trick: Slide a pencil under one side of the rubber bands to lift them like little guitar strings. This usually makes the sounds easier to hear.

Step 3 — Tight and Loose Test (8 min)

Gently pull one rubber band tighter and pluck it. Then let it relax and pluck it again. Ask: *"Did the tighter band sound higher or lower?"*

Children should notice that tighter often sounds higher, while looser often sounds lower.

Step 4 — Short and Long Test (8 min)

If you have a ruler, place it on a table with one end hanging off. Pluck the hanging end. Try a short overhang and then a longer overhang. Talk about which sound is higher and which is lower.



Higher Sound

Often comes from a **faster** vibration, a **tighter** band, or a **shorter** vibrating part.



Lower Sound

Often comes from a **slower** vibration, a **looser** band, or a **longer** vibrating part.

Step 5 — Try Your Own Voice (5 min)

Say "hello" in a high voice and then in a low voice. Ask: "How can our voices make different sounds too?" Connect this back to the idea that our vocal cords vibrate to make sound.



Discussion Questions



Question 1: "How was this lesson different from Lesson 1?"



In Lesson 1, we learned that sound is made by vibration. In this lesson, we compare different sounds and notice that vibrations can make sounds that are higher or lower.



Question 2: "What change made the sound higher?"



Strong answers might include making the rubber band tighter, using a thinner band, or making the vibrating part shorter.



Question 3: "What made the sound lower?"



Strong answers might include using a thicker or looser band, or making the vibrating part longer.



Extensions

- 🎵 Make a pattern like high-high-low or low-high-low and repeat it.
- 🐦 Listen for high bird sounds outside and compare them to a lower grown-up voice.
- 🥁 Compare the sound of a small container and a larger container when tapped gently.



Parent/Teacher Notes

- **This lesson should feel meaningfully different from Lesson 1:** Lesson 1 establishes that sound comes from vibration. Lesson 4 deepens that idea by helping students compare pitch — higher and lower sounds — using the same vibration concept.
- **Keep the language simple:** Grade 1 students do not need formal frequency vocabulary, but hearing "faster vibration" and "slower vibration" is appropriate.
- **Success bar:** students can name a sound as high or low and connect that difference to how the object was vibrating.



Student Worksheet — High Sounds, Low Sounds

Name: _____ Date: _____

Part 1 — What Did You Hear?

Test	Was it higher or lower?	What changed?
Thin band / thick band		
Tight band / loose band		
Short ruler / long ruler		

Part 2 — Finish the Sentences

A higher sound often comes from a _____ vibration.

A lower sound often comes from a _____ vibration.

Lesson 1 taught me that sound is made by _____.

Part 3 — Draw a High Sound and a Low Sound

Draw two sound makers. Label one higher and one lower.

Draw here 



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1

Answers may vary slightly, but typical responses are:

- **Thin band** = higher, **thick band** = lower
- **Tight band** = higher, **loose band** = lower
- **Short ruler overhang** = higher, **long ruler overhang** = lower

Part 2

- A higher sound often comes from a **faster** vibration.
- A lower sound often comes from a **slower** vibration.
- Lesson 1 taught me that sound is made by **vibrations**.

Part 3

Any sensible drawing of two different sound makers, or one sound maker making a high and low sound, is acceptable.



What Success Looks Like

- Child can describe one sound as higher and another as lower.
- Child can explain that both sounds still came from vibrations.
- Child can point to a change that helped make the sound higher or lower.

Sound Through Solids, Liquids & Gases

Physical Science — Sound & Light | Unit 1, Lesson 5 of 6

 **Grade:** 1st Grade

 **Duration:** 30–45 minutes

 **Subject:** Physical Science — Sound & Light



Sound can travel through many kinds of matter — not just air.

NGSS Standards:

1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

Learning Objective

Students will notice that **sound can travel through different materials** and explain that **sound is made by vibrations**.

The Big Idea

Sound needs matter to travel. It can move through air, through water, and through solid objects like tables, walls, spoons, and string. In every case, something starts vibrating, and that vibration moves through the material until it reaches our ears.

 **Key vocabulary:** sound, vibrate, solid, liquid, gas, air, matter, travel



Materials



Gather Before You Start

- Plastic cup or bowl of water
- Metal spoon
- Wooden table or desk
- Optional: string and 2 cups for a cup phone



For the Worksheet

- Printed Student Worksheet
- Pencil or crayons



Let's Get Started

Step 1 — Sound Through Air (5 min)

Tap a spoon gently on a cup while your child listens from nearby. Ask: *"What is the sound traveling through right now?"* Help them name the air around them. Then ask: *"What was moving or shaking to make the sound?"* Help them notice that the spoon or cup was vibrating.

Step 2 — Sound Through a Table (8 min)

Have your child put one ear gently on a wooden table. Tap the table lightly with your finger or spoon a short distance away. Then listen again with the ear off the table.

Ask: *"Did the sound seem the same or different?"*



What to notice: The table is a solid. When you tap it, the table vibrates, and that vibration can travel through the solid table as well as through the air.

Step 3 — Sound Through Water (8 min)

Tap the outside of a bowl or cup of water. Have your child listen beside the bowl while you tap gently. Talk about how the sound is still moving through matter — now including liquid water.

Step 4 — Spoon String Sound Test (10 min)

Tie a piece of string around the handle of a metal spoon. Hold the loose ends of the string and gently swing the spoon so it taps the edge of a table or chair. Listen to the sound with the string hanging free. Then wrap the string ends around one finger on each hand and press those fingers gently against the small flap in front of your ears while the spoon taps again.

Ask: *"When did the sound seem louder or clearer?"* and *"How did the string help the vibration travel to your ears?"*



Gas

Air is a gas. Sound can travel through air.



Liquid

Water is a liquid. Sound can travel through water.



Solid

Tables, walls, spoons, and string are solids. Sound can travel through solids.



Discussion Questions



Question 1: "What does sound need in order to travel?"



Sound needs matter — like air, water, string, or wood — because sound is a vibration moving through something. Ask: "What did we test that helped prove that?"



Question 2: "Why could you hear the tap through the table?"



*Because the table is a solid, and the **vibration** traveled through the solid material to the ear.*



Question 3: "What changed when the spoon sound traveled through the string to your ears?"



The sound usually seemed louder or clearer because the vibration had a direct path through the solid string to the ears.



Extensions

-  Listen at a door or wall when someone taps on the other side.
-  Try the spoon-string sound test with a wooden spoon and compare it to a metal spoon.
-  Notice real life examples: hearing splashing in a bathtub, hearing footsteps through a floor, hearing voices through a closed door.



Parent/Teacher Notes

- **Keep it concrete:** Grade 1 students do not need vacuum/space explanations here. Just help them notice that sound travels through different materials.
- **Best assessment:** Can the child name at least two materials sound traveled through in this lesson?
- **Safety:** Use gentle taps only. No loud banging near ears.



Student Worksheet — Sound Through Solids, Liquids & Gases

Name: _____ Date: _____

Part 1 — What Did You Test?

Test	What was the sound traveling through?	What did I notice?
Tap heard through table		
Tap heard near water bowl		
Spoon sound through string		

Part 2 — Finish the Sentences

Sound can travel through air, water, and _____.

One solid we tested was _____.

Sound is made by _____.

Part 3 — Draw One Example

Draw one sound traveling through matter.

Draw here 



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1

Accept answers such as: table/wood, water, string, air. Observations may vary. Many children notice the table tap sounds louder or clearer with an ear on the table than with their ear off the table. In the spoon-string test, many children notice the tapping sound seems much louder or clearer when the string helps carry the vibration directly to their ears.

Part 2

- Sound can travel through air, water, and **solids**.
- One solid we tested was **a table, string**, or another tested solid.
- Sound is made by **vibrations**.

Part 3

Any sensible drawing showing sound moving through a material is correct.



What Success Looks Like

- Child can name at least two materials sound traveled through.
- Child can explain that sound is vibration moving through matter.



Sound and Light Message Challenge



Physical Science — Sound & Light | Unit 1, Lesson 6 of 6

 **Grade:** 1st Grade

 **Duration:** 35–45 minutes

 **Subject:** Physical Science — Sound & Light



Sound and light can both carry messages, but each works best in different situations.

NGSS Standards:

1-PS4-1 — Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

Learning Objective

Students will **PLAN**, **TEST**, and **COMPARE** a sound signal and a light signal, then explain which one works best for a communication problem and why.



The Big Idea

This culminating lesson brings the whole unit together. Scientists learned that **sound starts with vibration** and can travel through matter. They also learned that **light travels in straight lines** and can be reflected with shiny surfaces. In this lesson, students use both ideas to solve a real communication problem: when should we use sound, and when should we use light?



Key vocabulary: communicate, signal, vibration, reflect, light, sound, compare, evidence



Materials



Gather Before You Start

- 1 flashlight
- 1 small mirror or a square of smooth aluminum foil on cardboard
- 1 metal spoon and 1 cup, pan, or table for making a sound signal
- 2 paper cards labeled **YES** and **NO** (optional)
- A pillow, closed door, or another simple obstacle for one test



For the Worksheet

- Printed Student Worksheet
- Pencil or crayons



Let's Get Started

Step 1 — Set the Problem (3 min)

Tell your child: *"Today you need to send a message to someone far away. Sometimes sound is the better tool. Sometimes light is the better tool. We are going to test both and decide."* Review the unit ideas: sound is vibration, and light travels in straight lines.

Step 2 — Make a Sound Signal (8 min)

Choose a simple sound signal, such as one tap for **yes** and two taps for **no**. Have one person tap a spoon on a cup, pan, or table while the partner listens from across the room. Then try again with the listener behind a door or pillow barrier. Ask: *"Could you still hear the message? What was vibrating?"*



Science connection: The spoon, cup, table, and air all help carry the vibration. The sound message works because something vibrates and that vibration travels to the listener.

Step 3 — Make a Light Signal (8 min)

Now choose a light signal, such as one flash for **yes** and two flashes for **no**. Shine a flashlight or bounce light from a mirror/foil toward a wall near the partner. Let the partner watch for the flashes and decode the message. Then test again with a barrier blocking the view. Ask: *"Did the light still work when something blocked the path?"*

Signal Detective:

Think about the evidence from both tests.

- Did the signal need a clear path?
- Could it go around a corner or through a door?
- Which signal was easier to notice?

Step 4 — Solve Three Mini Problems (10 min)

Read each situation and let your child choose **sound** or **light**.

- **Problem A:** Your friend is in another room and cannot see you.
- **Problem B:** Your friend is across the yard and can see your flashlight.
- **Problem C:** It is dark, and you want someone to notice where you are.

After each choice, ask: "What evidence from our tests helps you decide?"

Step 5 — Share the Best Tool (5 min)

Have your child explain which kind of signal was best for each problem. Encourage them to use the words **vibration**, **light**, **blocked**, and **travel**.

Discussion Questions

 **Question 1:** "When was sound the better signal?"

 *Sound can be better when the other person cannot see you, such as behind a door or around a corner.*

 **Question 2:** "When was light the better signal?"

 *Light can be better when the other person is far away and has a clear line of sight to the signal.*

 **Question 3:** "Why do sound and light not always work the same way?"

 *Sound is vibration traveling through matter. Light travels in straight lines and can be blocked if something is in the way.*

Extensions

-  Invent a new code with three messages, such as **come here**, **wait**, and **help**.
-  Try the light signal outdoors at dusk with adult help.
-  Test whether a thicker door or pillow changes how well the sound signal works.



Parent/Teacher Notes

- **This lesson is the unit capstone:** students are not building another cup phone. They are comparing **two different ways** to communicate and choosing the best tool for each situation.
- **Ideal explanation:** "I chose sound when my partner could not see me, and I chose light when my partner could see the flashes from far away."
- **Success bar:** students should use evidence from the tests, not just guess their favorite signal.



Student Worksheet — Sound and Light Message Challenge

Name: _____ Date: _____

Part 1 — Test Two Signals

Signal	How did we send it?	Did the message work?	What did I notice?
Sound			
Light			

Part 2 — Pick the Best Tool

Circle the better tool for each problem.

Problem	Better Tool	Why?
My friend is behind a door.	Sound / Light	
My friend is far away and can see me.	Sound / Light	
I want someone to notice me in the dark.	Sound / Light	

Part 3 — Finish the Sentences

Sound works because something is _____.

Light works best when it has a clear _____.

The best signal for a hidden partner is _____ because _____.



PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Part 1

Accept sensible notes about each test. Strong answers mention that the sound signal could still work when the partner could not see, while the light signal needed a clear path.

Part 2

- **Behind a door:** usually **Sound**
- **Far away and can see me:** usually **Light**
- **Notice me in the dark:** often **Light**, though some children may defend sound with evidence; accept well-supported answers.

Part 3

- Sound works because something is **vibrating**.
- Light works best when it has a clear **path**.
- The best signal for a hidden partner is **sound** because **the partner can hear it even if they cannot see me**.



What Success Looks Like

- Child compares both kinds of signals.
- Child uses evidence from tests to match a signal to a problem.
- Child explains the choice using vibration and straight-line light ideas from the unit.



Grade 1 Sound & Light Reference Card

VIBRATION • PITCH • SOUND TRAVEL • LIGHT



SOUND

Sound is made by vibrations

If the vibration stops, the sound stops

We can sometimes see objects shake

Drums, rulers, and strings can vibrate



PITCH

Pitch means high or low sound

Faster vibration often sounds higher

Slower vibration often sounds lower

Tight or short bands often sound higher



SOUND TRAVEL

Sound can travel through air



LIGHT

Light travels in straight lines



NO VIBRATION

No sound is being made



LOOSE STRING

A sound sender works less clearly



BLOCKED LIGHT

Light cannot pass through



NO MATTER

Sound cannot travel through empty space



Thin or tight strings often make higher sounds



Cup phones work best when the string is tight



A table can carry sound because it is a solid



Mirrors help light bounce where we want it to go



Sound & Light Quiz



Physical Science · Unit 1 · Grade 1

QUESTION 1 OF 10



What causes ALL sounds?

Moving air through a fan

Vibrations — things moving back and forth very fast

Electricity flowing through a wire

Heat from the sun

QUESTION 2 OF 10



You pluck a rubber band and it makes a sound. Then you press it to stop it from moving. What happens to the sound?

It gets louder

It keeps going for a while

It stops immediately

It changes pitch

QUESTION 3 OF 10



You have two rubber bands — one thick and one thin. When you pluck them, which one makes a HIGHER pitch sound?

The thick rubber band

The thin rubber band

They both sound the same

Pitch has nothing to do with thickness

QUESTION 4 OF 10



You bang a drum SOFTLY, then HARD. What changes about the sound?

The pitch gets higher

The sound gets louder (more volume)

The sound gets faster

Nothing changes

QUESTION 5 OF 10



You shine a flashlight across a dark room. How does the light travel?

In a curved path around objects

In a straight line

It bounces in all directions at once

It swirls like water

QUESTION 6 OF 10



You hold plastic wrap in front of a flashlight. Light passes through clearly and you can see objects on the other side. What word describes plastic wrap?

Opaque

Translucent

Transparent

Reflective

QUESTION 7 OF 10



You hold wax paper in front of a flashlight. Some light gets through, but everything looks blurry — not clear. What word describes wax paper?

Transparent

Translucent

Opaque

Reflective

QUESTION 8 OF 10



You hold a piece of cardboard in front of a flashlight. No light gets through at all — a dark shadow appears behind it. What word describes cardboard?

Transparent

Translucent

Opaque

Luminous

QUESTION 9 OF 10



You use a mirror to bounce a beam of light onto a different spot on the wall. What is this called?

Translucent

Absorption

Reflection

Vibration

QUESTION 10 OF 10



People have used light and sound to send messages for thousands of years. Which TWO things below are examples of using light or sound to communicate across a distance?

Flashing a mirror signal ♦ Drum code beats

Writing a letter ♦ Drawing a picture

Building a fire to cook food ♦ Humming quietly to yourself

Eating lunch ♦ Taking a nap

Sound & Light Unit Quiz Answer Key

Little Lab Coats • Grade 1 • Parent/Teacher Copy

1. 🎵 What causes ALL sounds?

Answer: B. Vibrations — things moving back and forth very fast

Every sound is made by vibrations! When something vibrates (moves quickly back and forth), it creates sound waves in the air that travel to your ears.

2. 🎸 You pluck a rubber band and it makes a sound. Then you press it to stop it from moving. What happens to the sound?

Answer: C. It stops immediately

The sound stops immediately! When you stop the vibration, you stop the sound. No vibration = no sound. That's the proof that sound is caused by vibrations.

3. 🎸 You have two rubber bands — one thick and one thin. When you pluck them, which one makes a HIGHER pitch sound?

Answer: B. The thin rubber band

The thin rubber band makes a higher pitch! Thin bands vibrate faster than thick bands. Faster vibrations = higher pitch. This is how guitars make different notes!

4. 🥁 You bang a drum SOFTLY, then HARD. What changes about the sound?

Answer: B. The sound gets louder (more volume)

Hitting harder makes bigger vibrations, and bigger vibrations = louder sound! Volume (how loud or quiet) depends on how big the vibrations are.

5. 🔦 You shine a flashlight across a dark room. How does the light travel?

Answer: B. In a straight line

Light always travels in a straight line — that's one of its most important properties! It cannot bend around corners. That's why you can't see around a wall.

6. 🪄 You hold plastic wrap in front of a flashlight. Light passes through clearly and you can see objects on the other side. What word describes plastic wrap?

Answer: C. Transparent

Transparent means ALL light passes through and you can see clearly! Examples: clear glass, plastic wrap, a clear window.

7.  You hold wax paper in front of a flashlight. Some light gets through, but everything looks blurry — not clear. What word describes wax paper?

Answer: B. Translucent

Translucent means SOME light gets through but it's blurry — you can't see clearly. Examples: wax paper, tissue paper, frosted glass.

8.  You hold a piece of cardboard in front of a flashlight. No light gets through at all — a dark shadow appears behind it. What word describes cardboard?

Answer: C. Opaque

Opaque means light is BLOCKED completely — you can't see through it at all. Examples: cardboard, wood, your hand, aluminum foil.

9.  You use a mirror to bounce a beam of light onto a different spot on the wall. What is this called?

Answer: C. Reflection

Reflection! When light hits a shiny surface like a mirror, it bounces (reflects) in a new direction. Sailors and hikers have used mirror reflections to send signals over long distances!

10.  People have used light and sound to send messages for thousands of years. Which TWO things below are examples of using light or sound to communicate across a distance?

Answer: A. Flashing a mirror signal ✦ Drum code beats

Flashing mirrors use reflected light to send signals! Drum codes use sound vibrations to carry messages far away. Humans have used both for thousands of years — long before phones existed!